

A per-seed no-failure certificate for SSS

How the proportional demand $\lambda\phi$ lets us prove safety that SSA cannot

SPARK goal-insertion notes

Setup

This note builds on `cx_derivation.pdf` (the ceiling $c(x)$) and `cx_policies.pdf` (the demand d per controller). Reuse all symbols from there:

symbol	meaning
$\phi_i(x)$	safety index of constraint i ; safe $\{\phi_i \leq 0\}$, danger $\{\phi_i > 0\}$
$\dot{\phi}_i = L_f\phi_i + L_g\phi_i u$	rate of change; here $f \equiv 0 \Rightarrow L_f\phi_i = 0$
$c_i(x) = \sum_k u_{lim,k} (L_g\phi_i)_k $	control-authority ceiling: fastest the motors can push ϕ_i down
η	SSA's constant demand: enforce $\dot{\phi}_i \leq -\eta$ when $\phi_i \geq 0$
$\lambda\phi_i$	SSS's proportional demand: enforce $\dot{\phi}_i \leq -\lambda\phi_i$ when $\phi_i \geq 0$
Δt	control/integration time step

We use one fact from `cx_derivation` Step 6 repeatedly: as the control ranges over the box, the reachable values of $\dot{\phi}_i$ form the symmetric interval

$$\dot{\phi}_i \in [-c_i(x), +c_i(x)].$$

So $c_i(x)$ is both the fastest ϕ_i can be driven down and the fastest it can rise.

What “SSS failure” means. SSS is a hard QP: at an active constraint it demands $\dot{\phi}_i \leq -\lambda\phi_i$, with no slack. If the box cannot meet that — i.e. $g_i(x) := c_i(x) - \lambda\phi_i < 0$ — the QP is infeasible, SSS gives up and returns u_{ref} , and the body is no longer filtered. That give-up is the failure.

The recap: the SSA certificate

For SSA the demand is the constant η , so feasibility of constraint i is $c_i(x) \geq \eta$ (with $L_f\phi = 0$). Hence the clean per-seed certificate:

$(\text{SSA}) \quad \forall \text{ reachable active } x : c_i(x) \geq \eta \implies \text{no SSA failure.}$

The bar η is fixed and independent of Δt .

The direct mirror for SSS (same logical status)

Replace the constant η by the state-dependent demand $\lambda\phi_i(x)$. The feasibility test of cx_policies is $c_i(x) \geq \lambda\phi_i(x)$, i.e. $g_i(x) \geq 0$, so the literal analog is:

$$(\text{SSS, direct}) \quad \forall \text{ reachable active } x : c_i(x) \geq \lambda\phi_i(x) \implies \text{no SSS failure.}$$

This is exactly $\min_x g(x) \geq 0$ over the reachable set — the quantity the policy sweep already measures. It is a valid sufficient condition of the same form as SSA's.

Shared caveat. Both certificates quantify over reachable active states. Getting that set exactly is the hard part for either filter. Sampling rollouts (as the sweep does) yields an empirical certificate; a formal proof needs a sound over-approximation of the reachable set. SSS is no worse off than SSA here.

The SSS upgrade: a forward-invariance bootstrap

SSS can do something SSA cannot: instead of checking $c \geq \lambda\phi$ over the entire danger region, we only need it on a thin boundary shell, because a working SSS keeps ϕ small. Three ingredients.

(i) One-step overshoot is bounded by $c \Delta t$. Since $\dot{\phi}_i \leq c_i(x)$ for any box control (top of the reachable interval), a single step starting from a safe state $\phi_i(x_k) \leq 0$ can raise the index by at most

$$\phi_i(x_{k+1}) \leq \phi_i(x_k) + c_i(x_k) \Delta t \leq c_{\max} \Delta t, \quad c_{\max} := \sup_{\text{reach.}} c_i(x).$$

(ii) The feasible recovery shell has thickness c/λ . SSS is feasible wherever

$$\lambda\phi_i \leq c_i(x) \iff \phi_i \leq \frac{c_i(x)}{\lambda}.$$

(iii) Inside the active shell, ϕ only decreases. When active ($\phi_i \geq 0$) and feasible, SSS enforces $\dot{\phi}_i \leq -\lambda\phi_i \leq 0$. In discrete time, with $\lambda\Delta t < 1$,

$$\phi_i(x_{k+1}) \leq (1 - \lambda\Delta t) \phi_i(x_k) < \phi_i(x_k).$$

So ϕ_i never climbs while the filter is on; it can only rise during inactive steps ($\phi_i < 0$), and each such rise is bounded by (i).

Putting it together (the invariant). Suppose the system starts safe, $\phi_i(x_0) \leq 0$. By (iii) ϕ_i rises only on boundary-crossing steps, and by (i) each crossing lands at $\phi_i \leq c_{\max}\Delta t$; thereafter (iii) drives it back down. Hence the invariant

$$\phi_i(x_k) \leq c_{\max} \Delta t \quad \text{for all } k.$$

Feasibility (ii) is preserved as long as this ceiling sits inside the shell at the landing state, $c_{\max}\Delta t \leq c_i(x_{k+1})/\lambda$. A single conservative scalar condition guaranteeing that everywhere is:

$$\text{start safe } (\phi(x_0) \leq 0), \quad \lambda\Delta t < 1, \quad \lambda\Delta t c_{\max} \leq c_{\min} \implies \text{no SSS failure,}$$

with $c_{\min} := \inf_{\text{reachable active}} c_i(x) > 0$. In words: the demand-amplified one-step overshoot $\lambda\Delta t c_{\max}$ cannot exceed the available braking authority $c_{\min}$.

Why this is strictly stronger than SSA's bar

	SSA certificate	SSS certificate
condition	$c(x) \geq \eta$ everywhere reachable	$\lambda \Delta t c_{\max} \leq c_{\min}$ (start safe, small Δt)
the bar	fixed η , independent of Δt	scales with Δt
satisfiable by	needs every reachable boundary $c \geq \eta$	any seed with $c_{\min} > 0$, for small enough Δt
irreducible failure set	$\{c(x) < \eta\}$ — a fat shell	$\{c(x) = 0\}$ — controllability singularities only

Because the SSS bar scales with Δt , every seed whose reachable boundary avoids a controllability singularity ($c(x) > 0$ throughout) is certifiable failure-free for small enough Δt . SSA cannot say this: its $\{c < \eta\}$ shell stays whatever Δt is. As $\Delta t \rightarrow 0$, SSS's vulnerable set collapses to the measure-zero singularity set $\{c(x) = 0\}$ of `cx_derivation` Step 8 — the irreducible floor where no controller of any kind survives.

Consistency with the empirical sweep

At the simulator's fixed Δt , the sweep found SSS failures on seeds 7 and 18 (and only those, out of 12 valid seeds). In the language above, those are exactly the seeds where

$$\lambda \Delta t c_{\max} > c_{\min}$$

— the inserted goal G'_1 steers the closed loop so a single inactive-region step overshoots past the shell ($c_{\min}$ dips near 0 at that contact), and the next QP is infeasible. For the other valid seeds the inequality can hold, so they are certifiable. The certificate is therefore a real, checkable per-seed test, not a vacuous one.

How to discharge it numerically (per seed)

For a given seed, instrument the rollouts and record, over the reachable active states:

1. $c_{\min} = \min_t \min_{i \text{ active}} c_i(x_t)$ and $c_{\max} = \max_t \max_i c_i(x_t)$;
2. the time step Δt and the gain λ ;
3. (optionally) the realized worst one-step rise $\max_t (\phi_i(x_{t+1}) - \phi_i(x_t))_+$ to tighten the conservative $c_{\max} \Delta t$ overshoot bound.

Then report certified if $\lambda \Delta t c_{\max} \leq c_{\min}$ and $c_{\min} > 0$, else not certified (with the violation margin $\lambda \Delta t c_{\max} - c_{\min}$). This yields a hard “certified / not-certified” column alongside the empirical $\min_x g(x)$.

Summary

- The direct SSS certificate is the SSA one with $\eta \rightarrow \lambda \phi(x)$: $c(x) \geq \lambda \phi(x)$ (i.e. $g \geq 0$) over the reachable active set. Same status, same reachability caveat.

- The upgraded SSS certificate uses forward invariance — overshoot $\leq c\Delta t$, shell thickness c/λ , decrease inside — to reduce the check to a single scalar inequality $\lambda\Delta t c_{\max} \leq c_{\min}$.
- Unlike SSA's fixed bar η , the SSS bar shrinks with Δt , so the only irreducible obstruction is a controllability singularity $c(x) = 0$. The proportional demand is exactly what buys this.